



**UNIVERSITI TEKNOLOGI MARA
FACULTY OF COMPUTER SCIENCE & MATHEMATICS**



**ResolveX Complaint Management
AI Log File**

GROUP: D1CDCS1104B

GROUP MEMBERS

NO	STUDENT'S NAME	STUDENT'S ID
1	NUR AIN NASIHAN BINTI NAZARUDIN	2024421616
2	SAFIYA NAJWA BINTI AZMAN	2024416962
3	NUR KHIRUNNISYA BINTI KAMARZUKI	2024407062
4	NUR FAIZAH BINTI ABDULL HADI	2024404546

LECTURER NAME: NOR ASMA BINTI MOHD ZIN

Entry 1: Password Recovery and Reset Module

AI Tool Used: Gemini

1. The Intent

To develop a secure password recovery feature that allows users to reset their passwords when they forget their login credentials. The objective was to create a complete password reset workflow, including email verification, verification code validation, and password update functionality.

2. The Interaction

- Uploaded and explained the existing login and user authentication structure, including the database design used to store user account information.
- Requested assistance in implementing a password recovery process consisting of three stages: requesting a password reset, verifying a security code, and creating a new password.
- The AI reviewed the proposed workflow and suggested separating the process into multiple pages to improve security and maintainability. It also provided recommendations for validating user input and preventing unauthorized password changes.

3. The Output and Verification

- Generated PHP code for the forgot password module (forgot_password.php), allowing users to submit their registered email address to initiate the password reset process.
- Provided logic for verification code generation and validation in verify_code.php to ensure that only authorized users could proceed with the password reset process.
- Generated code for reset_password.php to allow users to create a new password after successful verification.
- Added validation checks to ensure that passwords met the required criteria and that incorrect verification codes were rejected.
- The generated code was reviewed and modified to match the project's database structure and user account table.
- The complete password recovery workflow was tested using multiple user accounts. Verification codes were successfully validated, password updates were stored correctly in the database, and users were able to log in using their newly created passwords.

Entry 2: Admin Complaint Management Interface

AI Tool Used: Gemini

1. The Intent:

To develop an efficient admin complaint management module that allows administrators to view complaint details, update complaint status, assign responsible staff, provide responses to complainants, and maintain a complete activity history. The objective was to create a structured and user-friendly interface that simplifies the complaint monitoring and management process.

2. The Interaction:

- Uploaded and explained the existing complaint management system structure, including the complaint database, admin dashboard workflow, and staff assignment information.
- Requested assistance in designing the admin complaint interface and implementing the update functionality, including viewing complaint details, changing complaint status, assigning staff, adding admin responses, and recording updates into the activity history.
- The AI analysed the system workflow and suggested organizing the interface into separate sections, such as complaint information, complaint update form, and activity history, to improve usability and make complaint management easier for administrators.

3. The Output and Verification:

AI provided interface design suggestions, PHP code, and SQL queries for the complaint management module. The generated output was reviewed and modified to match the existing system requirements and database structure. The functionality was tested to ensure that complaint information was displayed correctly, updates were saved successfully, and activity history was recorded accurately.

Entry 3: Complaint Submission Module

AI Tool Used: ChatGPT

1. The Intent:

To develop a complete complaint submission feature that allows users to report issues by providing complaint location details, selecting the type of damage, writing a complaint description, and uploading supporting evidence. The objective was to create a structured complaint submission workflow that ensures all required information is collected accurately and stored successfully in the system database.

2. The Interaction:

- Uploaded and explained the existing complaint submission interface, including the form structure, database design, and the required fields such as complaint area, location details, type of damage, complaint description, and evidence upload.
- Requested assistance in implementing the complaint submission process using PHP and MySQL, including form validation, file upload handling, complaint data insertion, and automatic generation of complaint IDs.
- The AI analyzed the proposed workflow and provided recommendations for organizing the form processing logic, validating user inputs, handling uploaded files securely, and ensuring the complaint information was stored correctly in the database.

3. The Output and Verification:

The AI provided suggestions for improving the complaint submission interface, including the arrangement of complaint area selection, location details, damage category, complaint description, evidence upload, and submission button to create a more organized and user-friendly layout.

The generated interface design was reviewed and adjusted according to the project requirements. The complaint submission page was tested to ensure that users could enter complaint information, upload evidence, and submit complaints successfully.

Entry 4 : Data Flow Diagram (DFD)

AI Tool Used : Microsoft Copilot

1. The Intent :

- The primary intent was to verify the accuracy of all arrows in the DFD. This ensures that data flows correctly between external entities, processes, and data stores, following standard DFD conventions. Proper arrow placement is critical because it reflects the logical movement of information within the system.
- Another intent was to identify and correct potential errors in the diagram. Common mistakes include arrows pointing directly from a data store to an external entity, or mislabelled arrows that do not clearly represent the type of data being transferred. Detecting these issues early helps maintain the integrity of the system model.
- Finally, the intent was to refine and expand the diagram by introducing subprocesses. Breaking down complex functions into smaller, well-defined subprocesses improves clarity, makes the system easier to understand, and ensures that each step of the complaint management workflow is properly represented.

2. The Interaction :

- The interaction involved a step-by-step review of the existing DFD, carefully checking each arrow's direction and ensuring logical consistency across all processes. This systematic approach allowed errors to be spotted and corrected effectively.
- During the review, feedback highlighted the need to add subprocesses to represent major system functions more clearly. For example, instead of showing a single "Complaint Handling" process, it was broken down into Complaint Submission, Complaint Review, and Complaint Update. This makes the diagram more precise and easier to interpret.
- The diagram was then reconstructed and improved based on these findings. Labels were corrected, subprocesses were added, and arrows were adjusted to comply with DFD standards. This interaction ensured that the final diagram was both technically accurate and academically suitable for inclusion in the project report.

3. The Output and Verification :

- The revised diagram confirmed that all arrows are correctly directed, eliminating invalid shortcuts between entities and data stores. This verification step ensures that the diagram accurately reflects the logical flow of information in the complaint management system.
- The DFD was expanded with subprocesses, including:
 - User Authentication – verifies login credentials for both students and admins.
 - Complaint Submission – allows students to submit complaints into the system.
 - Complaint Review – enables admins to review submitted complaints.
 - Complaint Update – updates complaint status in the database.
 - Generate Reports – produces complaint summaries and reports for admin viewing.
- Verification showed that the new structure provides a systematic and comprehensive representation of how data flows through the complaint management system. It highlights the interaction between entities, processes, and data stores, making the diagram suitable for academic documentation and professional presentation.